

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1328

Berrywood 100 kV
March 01, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1328. The Interconnection Customer proposes a 69.4 MW battery energy storage facility interconnecting at the Berrywood 100 kV substation in the Ohio area. The request was received on November 03, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 453 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$147,366,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mariposa – Acec Brooks 100.0 kV	Norich – Barrett 500.0 kV	177.9	150	7.3
Mariposa – Mosher 100.0 kV	Norwich Substation – Barrett 500.0 kV	175.7	140	7.3
Bishop Creek 3 – Clapham 100.0 kV line	Trego – Beaver 500.0 kV	162.5	139	5.0
Eustis Substation – Bishop Creek 3 100.0 kV Line	Trego Substation – Beaver 500.0 kV	161.9	136	5.0
Eustis – North Trail 100.0 kV line	Trego – Beaver Substation 500.0 kV	158.7	134	5.0
Sithe – Mechanicsburg 100.0 kV	Wolfe – Barrett 500.0 kV Line	157.4	141	5.4
Drumrock – Apple Vly 100.0 kV	Wolfe – Barrett 500.0 kV	152.5	121	6.5
Downs – Alps 100.0 kV	Barrett – Norwich 500.0 kV line	139.5	129	5.4
Carnal – Tupelo 100.0 kV	Wolfe – Barrett 500.0 kV	131.7	129	9.0
Downs Substation – Wilmington Sta C Tap 100.0 kV line	Norwich – Barrett Substation 500.0 kV	130.3	158	5.4
Drumrock – Hayden 100.0 kV	Wolfe – Barrett 500.0 kV	124.9	130	6.5
Wansley – Acec Brooks 100.0 kV	Wanley Substation – Acec Brooks 100.0 kV #1	124.3	193	9.3
Tupelo – Hyatt 100.0 kV L	Wolfe – Barrett 500.0 kV	123.5	119	9.0
Tejon – Neligh 100.0 kV	Apple Vly – N Bench 100.0 kV	120.8	109	5.4
Lincoln St W Substation – Tejon 100.0 kV	Billerica – Telephone 100.0 kV Line	119.9	112	100.0
Carley Road – Carnal 100.0 kV	Wolfe – Barrett 500.0 kV	119.7	205	5.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Floral Hills – Carley Rd 100.0 kV	Wolfe – Barrett 500.0 kV line	117.6	146	7.0
Wansley – Acec Brooks 100.0 kV L	Wansley Substation – Acec Brooks 100.0 kV #2	112.7	213	9.3
Floral Hills – Helena 100.0 kV L	Wolfe – Barrett 500.0 kV	110.6	144	7.0
Helena – Ductile 100.0 kV	Barrett Substation – Wolfe 500.0 kV L	108.0	141	7.0
Hyatt – Telephone 100.0 kV	Lincoln St West – Tejon 100.0 kV	107.9	146	63.1
North Trail – Uilk Wind Llc 100.0 kV Line	Trego – Beaver Substation 500.0 kV Line	106.3	126	9.4
Tejon Substation – Ductile Substation 100.0 kV Line	Wolfe – Barrett 500.0 kV line	104.4	139	7.0
Winthrop – Wanley 100.0 kV Line 1	Wansley – Winthrop 100.0 kV	104.1	164	11.3
Winthrop – Wansley 100.0 kV L	Wansley Substation – Winthrop 100.0 kV Line 1	104.1	164	11.3
Apple Vly – Sithe 100.0 kV	Carnal – Tupelo 100.0 kV	104.0	215	5.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mariposa – Acec Brooks 100.0 kV	Rebuild line with higher-capacity conductor	\$3,626,000	47% (\$1,707,000)
Mariposa – Mosher 100.0 kV	Rebuild line with higher-capacity conductor	\$2,347,000	47% (\$1,104,000)
Bishop Creek 3 – Clapham 100.0 kV	Rebuild line with higher-capacity conductor	\$12,026,000	100% (\$12,026,000)
Eustis – Bishop Crk 3 Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$9,053,000	100% (\$9,053,000)
Eustis – North Trail 100.0 kV line	Rebuild line with higher-capacity conductor	\$9,883,000	100% (\$9,883,000)
Sithe – Mechanicsburg 100.0 kV	Rebuild line with higher-capacity conductor	\$13,381,000	100% (\$13,381,000)
Drumrock – Apple Vly 100.0 kV	Rebuild line with higher-capacity conductor	\$3,115,000	36% (\$1,110,000)
Downs – Alps 100.0 kV Line	Rebuild line with higher-capacity conductor	\$918,000	100% (\$918,000)
Carnal – Tupelo 100.0 kV	Rebuild line with higher-capacity conductor	\$6,093,000	100% (\$6,093,000)
Downs – Wilmington Sta C Tap 100.0 kV	Rebuild line with higher-capacity conductor	\$1,093,000	100% (\$1,093,000)
Drumrock – Hayden 100.0 kV	Rebuild line with higher-capacity conductor	\$5,667,000	36% (\$2,020,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wansley – Acec Brooks 100.0 kV #2	Rebuild line with higher-capacity conductor	\$1,440,000	100% (\$1,440,000)
Tupelo – Hyatt 100.0 kV L	Rebuild line with higher-capacity conductor	\$20,476,000	100% (\$20,476,000)
Tejon – Neligh 100.0 kV	Rebuild line with higher-capacity conductor	\$3,302,000	100% (\$3,302,000)
Tejon – LincolnSt West 100.0 kV	Rebuild line with higher-capacity conductor	\$9,540,000	100% (\$9,540,000)
Carley Road – Carnal 100.0 kV	Rebuild line with higher-capacity conductor	\$17,204,000	24% (\$4,157,000)
Floral Hills – Carley Road 100.0 kV	Rebuild line with higher-capacity conductor	\$25,658,000	100% (\$25,658,000)
Wansley – Acec Brooks 100.0 kV Line 1	Rebuild line with higher-capacity conductor	\$2,595,000	100% (\$2,595,000)
Floral Hills – Helena 100.0 kV	Rebuild line with higher-capacity conductor	\$13,023,000	100% (\$13,023,000)
Ductile – Helena 100.0 kV	Reconductor line	\$1,419,000	100% (\$1,419,000)
Hyatt – Telephone 100.0 kV	Reconductor line	\$1,273,000	100% (\$1,273,000)
North Trail – Uilk Wind Llc 100.0 kV Line	Reconductor line	\$1,977,000	100% (\$1,977,000)
Ductile – Tejon 100.0 kV Line	Reconductor line	\$2,378,000	100% (\$2,378,000)
Winthrop – Wanley 100.0 kV Line 1	Reconductor line	\$458,000	100% (\$458,000)
Winthrop – Wansley 100.0 kV L	Reconductor line	\$841,000	100% (\$841,000)
Apple Vly – Sithe 100.0 kV	Reconductor line	\$1,491,000	30% (\$441,000)
	Total allocated to project		\$147,366,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.